Hardware & Networking

OCP Insights: Day 1 Highlights Increasing Complexity in Scaling AI Infrastructure

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We attended Day 1 of OCP 2026 in San Jose, California, where the overarching theme across the keynotes and executive sessions was the increasing complexity of scaling AI infrastructure, particularly with the rapid growth in inference and agentic workloads driving greater emphasis on system-level efficiency, networking, power delivery, and open architectures rather than simply adding accelerator capacity. Networking was particularly prominent, with discussions spanning scale-out, scale-across, and scale-up, including growing adoption of Ethernet in the latter and the need to interconnect increasingly large clusters across multiple data centers. Meanwhile, power availability and efficiency remain critical constraints, driving innovation around 800V DC architectures, liquid cooling, and software-based management to maximize compute capacity within existing power budgets. Connectivity was another major focus, with advancements across copper, NPO/CPO, and emerging optical technologies aimed at addressing increasing bandwidth density, reach, and reliability requirements. Underpinning these developments was a consistent emphasis on open standards, interoperability, and hardware/software co-design, alongside the growing importance of CPUs in supporting agentic AI workloads.*Takeaways from individual keynotes and executive sessions include*:

NVIDIA highlights breadth across compute, networking, racks and factory solutions. Starting at the rack, Vera Rubin is now in full production, and its cable-free tray design has cut assembly time from ~90 minutes on Grace Blackwell to ~1 minute, supporting output of thousands of racks per week. In relation to networking, Spectrum-X Photonics is NVIDIA's first CPO Ethernet switch, which delivers ~5x higher power efficiency and adds multi-plane support with hardware-based link-fault detection, with shipments expected this year. Moving up to the facility, NVIDIA showcased the first 800V DC AI factory built to the emerging OCP standard, which uses ~45% less copper and scales from ~100kW to 1MW racks. NVIDIA is also contributing its DSX reference design, a grid-flexibility specification developed with Google, Meta and Microsoft, and open-source power-management software that taps unused provisioned power to fit more GPUs within the same envelope.

Networking takes center stage in Meta's AI scaling roadmap. Meta reinforced the growing importance of networking and full-stack infrastructure optimization in enabling AI scaling. Management highlighted lessons from its 1 GW Prometheus training cluster, which spans multiple buildings, as well as its planned 5 GW Hyperion site. On scale-out, Meta introduced its liquid-cooled IcePack platform, incorporating 102.4T IceCube switches that leverage silicon from Broadcom and Nvidia, with system partners such as Celestica and Accton highlighted. Meta also introduced its Flattened Planar Fabric (FPF), a new eight-plane, two-tier architecture that reduces latency, network complexity, and switching requirements. Meta also noted that it is working with third-party CSPs to deploy the latter, as it relies on external compute capacity alongside its internal infrastructure. On scale-up, Meta confirmed its adoption of Ethernet with MTIA, leveraging Broadcom, and noted the eventual adoption of the ESUN standard. On scale-across, Meta highlighted the challenges of distributed training and pointed to Ethernet back-end aggregation networks, which connect data centers through a

See page 8 for analyst certification and important disclosures.

super-spine architecture, leveraging deep buffering to address latency and lossless-network requirements. That said, the introduction of a new regional backbone was highlighted leveraging FBOSS, OCP SAI, and shallow-buffer switches, with Cisco Silicon One noted as the silicon of choice and production expected in five months. Separately, Meta announced the open sourcing of Meta RoCE, its RDMA transport protocol.

Broadcom points to scale-up Ethernet momentum, copper runway, and rising optical inter-rack needs. Broadcom highlighted its expectation that Ethernet will serve as the common networking foundation across scale-out, scale-across, and scale-up infrastructure, with increasing adoption of the latter representing a meaningful opportunity for its switching portfolio. Importantly, 5+ hyperscalers are already deploying Ethernet scale-up in production, with 10+ ecosystem partners developing solutions, underscoring accelerating adoption supported by open standards, including ESUN and SUE-T. Broadcom emphasized Ethernet's ability to deliver the low latency, reliability, and bandwidth efficiency required for tightly coupled XPU clusters, while leveraging common networking infrastructure and operational tools across all three domains. On the roadmap, Broadcom noted further runway for copper, outlining plans to double bandwidth over the next two years while maintaining 2 m reach, potentially extending copper's relevance within racks even as bandwidth requirements increase. Meanwhile, Broadcom highlighted opportunities to expand scale-up architectures toward ~1 k XPUs through networking and software optimization, with optical becoming increasingly important as scale-up connectivity extends across racks.

AMD highlighted robust compute demand and introduced Agentic AI rack platform - Helion. CEO Dr. Lisa Su highlighted that data center AI accelerator TAM is expected to grow at a >45% CAGR from ~$200 bn in 2025 to ~$1.4 tn by 2030, with inference now the #1 AI workload due to accelerating agentic adoption (agentic token usage up >14x YTD). Agentic workloads are also driving a sharp step-up in the data center CPU TAM, which is now expected to grow at a >50% CAGR from ~$26 bn in 2025 to ~$220 bn by 2030. Looking ahead, AMD sees agentic workloads requiring distinct rack architectures, with hyperscalers likely to deploy disaggregated CPU racks alongside GPU racks, while enterprises running on-prem may prefer all-in-one racks combining CPUs and GPUs optimized for their workloads. To address both, AMD announced a new Helion Agentic AI Rack, an OCP-compliant ORv3 rack built on standard modular compute boards that can be configured as a CPU-only rack deployed alongside GPU racks in hyperscale environments, or as an all-in-one CPU/GPU rack for enterprises. The rack includes CPUs, GPUs, networking and an integrated token router. Separately, management highlighted that shipments for Helios racks began in September, with the first systems installed at OpenAI, Meta, Anthropic, Microsoft and Oracle, with further ramp into 4Q and beyond.

ARM highlighted three distinct CPU use cases in data centers to support a $300 bn TAM by 2030. Breaking down the roles, AI head nodes feed the accelerator by scheduling, moving data and coordinating XPUs so costly accelerators never sit idle. Cloud-native CPUs, paired with DPUs for isolation, run the orchestration, tools and data services for every human user. Agentic CPUs are a new category, as each agent operates as a virtual user requiring its own dedicated, isolated VM. Together, these use cases drive a TAM expanding from $52 bn in 2026 to $300 bn by 2030 (~55% CAGR).

Amphenol highlighted continued innovation in copper as well as the need for co-development of copper and optics. On the copper roadmap, 400G per lane is viewed as feasible, with a 448G front I/O study showing a usable channel out to ~110GHz. Several customers continue to pursue PAM6 on the backplane even as optics converge on PAM4. Whether copper or optical, linear ports are defined by electrical channel performance, which makes co-developing the two more valuable than treating them as an either/or choice. Additionally, the company highlighted modularity in development, including the use of the

same base socket for both NPC and NPO solutions. Since the main source of copper signal loss is the chip package and host PCB rather than the cable cartridge, launching twinax directly from the package (CPC) and bypassing the PCB frees enough loss budget to extend copper reach to ~3.8m, enabling copper scale-up across two racks.

TeraHop highlighted three MSAs that address AI scaling bottlenecks across bandwidth capacity, density, power and fiber infrastructure. Firstly, the XPO MSA defines a 12.8T pluggable delivering 8x the capacity of a 1.6T module and 4x the faceplate density, with the industry's first integrated liquid cooled pluggable optical module. Secondly, the Open CPX MSA standardizes connectors and optical engines with integrated or external laser options and two connector types, which can be deployed as NPO, CPO or IOA (*OE sits on an HDI board attached to the substrate via a high-density elastomer connector*). TeraHop's 6.4T "Diablo-1" engine uses an integrated laser and achieves ~5.5pJ/b including the laser, vs. external laser approaches, where light passing through additional couplings, PM fiber and blind-mate connectors loses over half its power. That loss requires lasers running at more than 2x the output, with ~4x the power dissipation and limited field reliability data. Thirdly, the SDM4 MCF MSA defines a four-core fiber that cuts fiber terminations by 4x and reduces cable size, weight and pathway requirements.

TE Connectivity discussed solutions across 800V rack architecture and passive optical path. On the power opportunity, TE is participating in both paths emerging from the 800V sidecar, including 1) where sidecar feeds a 50V IT rack, and TE is prototyping liquid-cooled busbars at 750kW to 1MW+, as well as high-current power shelf connectors; and 2) where 800V is carried directly into the IT rack, and TE is building 800V IT rack busbars scaling from 250kW to 1.6MW alongside high-voltage busbar connectors. As power conversion is pushed further downstream, TE is developing a 1OU rack-to-rack interconnect solution to move power between racks. In relation to optics, TE is addressing the passive optical path across chip to system through three layers: 1) chip-to-fiber attach through fiber array units (FAUs) coupling high-density 1D and 2D fiber arrays to photonic engines; 2) fiber-to-fiber connectors and adapters at the faceplate, enabling low-loss, repeatable disconnect and reconnect, partly through partnerships with Senko, US Conec and 3M; and 3) system interconnect, including fanouts, backplanes, shuffles and large fiber assemblies routing fibers across trays and racks.

Credo pushes beyond copper with ALCs, including ecosystem support. Credo CEO Bill Brennan outlined the company's next-gen optical roadmap leveraging micro-emitter technology, including both microLEDs and VCSELs, which is expected to overcome the reach, bandwidth-density, and reliability limitations of copper. This approach leverages hundreds of lower-speed optical lanes with built-in redundancy to enable hitless failover, delivering copper-like reliability while consuming <2pJ/bit versus 5pJ/bit for passive copper-based architectures, representing power savings of >60%. The first commercial micro-emitter product, a 1.6T Active Light Cable (ALC), is targeted for production in 2027, offering 50m reach, 10x thinner and lighter cabling, and 30% lower power consumption. Additionally, a 3.2T ALC, 6.4T NPO, and prototype optical backplane were highlighted, underscoring a strong future roadmap, and supported by an announced industry workstream involving an unnamed hyperscaler and two major GPU manufacturers to establish a broader ecosystem.

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Announcements

Arista

- **Launches Etherlink SU-144, an Ethernet-based scale-up architecture built on liquid-cooled rack designs** (press release). The system connects up to 144 accelerators in a single network hop and up to 1,024 across racks. Arista announced three form factors: an orthogonal single-rack chassis with 100–400kW of liquid cooling for extreme bandwidth and density in intensive AI training and inference workloads, a cabled-backplane rack built for signal quality and easy servicing, and a modular, standalone cross-rack fabric that expands across multiple racks.
- **Introduces rack-scale open Ethernet scale-out networking designs that work with a variety of scale-up protocols** (press release). The racks support ESUN, NVLink, NVLink Fusion or PCIe and run either EOS or an open network operating system. Each rack includes several liquid-cooled 102.4T 7060EX7 switches, fiber patch panels, cooling manifolds, leak-detecting drip trays, rack management controllers, power shelves and battery backup units.
- **Microsoft and d-Matrix share scale-up engagements with Arista; Meta and AMD appear limited to scale-out** (press release). Microsoft is extending its partnership with Arista to design a scale-up solution for its Maia accelerators, describing it as a co-engineered compute and network rack. d-Matrix is building an ESUN rack that connects 128 next-gen Raptor XPUs with Arista switches for low-latency inference. Meta is collaborating with Arista on rack-scale networking solutions for Meta's 102.4T scale-out platforms. Lastly, AMD's partnership with Arista entails pairing Arista switches (*scale-out*) with its Vulcano 800 AI NICs.

Broadcom

- **Highlights its networking and integrated optics portfolio for AI scale-up, scale-out, and scale-across architectures** (press release). Broadcom's showcase includes Tomahawk 6, Tomahawk Ultra, and Jericho 4 Ethernet switches, 800G and 400G AI NICs, and third-generation TH6-Davisson co-packaged optics. The portfolio supports ORV3 rack-based solutions, with Broadcom emphasizing bandwidth, power efficiency, and modular designs for increasingly large AI clusters.

Cisco

- **P200-based systems are now in production at hyperscalers, with Microsoft and Alibaba named** (press release). Cisco claims to be the first vendor to announce 51.2T systems built on Silicon One P200 running in live production at leading hyperscalers. Microsoft has moved P200-based systems into production and Alibaba has deployed its in-house Quantum Cat router, built on P200 silicon, at massive scale for its eCORE backbone, pairing Cisco's silicon with Alibaba's custom systems and SONiC.
- **Cisco and NVIDIA introduce the Ultra-Dense Small Form Factor Pluggable (USFP), a pluggable optics form factor that runs 400G per lane** (press release). The first planned transceivers carry 16 full-duplex lanes for switches and SuperNICs and can grow to 32 lanes or more. The companies claim 4x the bandwidth density and 1.3x the power efficiency of 200G-per-lane optics. They say it supports up to 100W of power dissipation without a dedicated embedded cooling circuit. They plan an open spec through an upcoming community-led multi-source project and are working with TE Connectivity and Amphenol on cages, connectors and modules.
- **Cisco's 102.4T G300 is now sampling with first systems shipping; adds the G302 for**

top-of-rack (press release). The 102.4T Silicon One G300, announced earlier this year, is now available for customer sampling, and the first G300-powered systems are shipping to customers. Cisco also introduced the Silicon One G302, which it calls the first 51.2T switching silicon with 200G SerDes designed for ToR deployments, with Cisco 1.6T optics and direct-to-chip liquid cooling integration. The G302 powers rack-level ToR switches that aggregate server traffic into the wider data center network and is planned to begin shipping before the end of 2026.

- **Adds Live Protect to Cisco-validated SONiC and brings SONiC to management networks** (press release). Customers can run Cisco-validated SONiC with Nexus One, which Cisco describes as an open approach for multi-site to rack-scale environments. Starting mid-2027, Live Protect will reduce customers' exposure between a vulnerability disclosure and a patch, giving them time to test and deploy the permanent fix without disrupting operations. Cisco also launched the Cisco 8001, a Silicon One A100-based management system available in 10G or 1G versions that brings SONiC to the management network.

Celestica

- **Introduces the DS6005, a fully liquid-cooled 102.4T co-packaged optics switch built on Broadcom's Tomahawk 6** (press release). The DS6005 offers up to 102.4Tbps of switching in a 1.5OU box sized for OCP ORv3 racks, with 64 x 1.6TbE ports. The switch was designed for both AI/ML scale-up and scale-out networking. Because the optical engines sit in the same package as the switch chip, the DS6005 does not need standard pluggable optics in the front panel. The design combines co-packaged optics, liquid cooling, external laser modules that can be swapped in the field, and management software built for co-packaged optics. It runs SONiC.

Credo

- **Announces its newest optical and copper interconnect products for scaling AI infrastructure** (press release). Highlights include 1.6T ZeroFlap active light cables, optical transceivers, and active electrical cables, alongside Toucan Gen6 PCIe AECs for rack-level connectivity. Credo positions its active light cables to address the mid-reach gap between copper's distance limitations and traditional pluggable optics' power and cost overhead. The showcase also included a SENKO-developed micro-emitter optical backplane delivering up to 10x higher bandwidth than the latest copper backplanes, integrated 1.6T DSP/silicon photonics modules, and PILOT-enabled telemetry and diagnostics across several offerings.
- **Launches an OCP workstream to standardize short-reach micro-emitter optics for AI data centers** (press release). The Short-Reach Micro-Emitter Optics (SRMO) initiative brings together a leading hyperscaler and optical connectivity providers, including ams OSRAM, SENKO, and others, to develop interoperable specifications for wide, parallel optical interconnects across scale-up and scale-out fabrics. The approach uses many micro-emitters operating in parallel to improve reliability, power efficiency, and bandwidth density. The specification is targeted for OCP approval in 2027, with requirements spanning signaling, power, form factors, connectors, and compliance.

HPE

- **Launches the ProLiant XD245 and XD285, two OCP-compliant servers built on 6th Gen AMD EPYC** (press release). The XD245 fits four direct liquid-cooled nodes into one Open Rack Unit (1OU) and uses up to eight 6th Gen EPYC processors. The XD285 is an air-cooled two-node design in two Open Rack Units (2OU), which HPE says brings the same design to more kinds of data centers. Both are aimed at CPU-heavy work such

as engineering simulation, weather and climate modeling, scientific research and financial modeling.

Keysight

- **Introduces two 1.6T Ethernet test platforms, the SERT 1600GE and the AresONE Plus 1600GE** (press release). The SERT 1600GE tests data center switches and edge routers. The AresONE Plus 1600GE is the first product to fit eight 1.6T Ethernet ports into one 2U box. The AresONE Plus also generates realistic AI traffic that tracks state and mimics a mix of collective operations, enabling engineers to check network utilization, congestion, load balancing and workload performance before they build large GPU clusters.

Lumentum

- **Announces a PCIe-over-optics demonstration with Astera Labs and Corning for rack-scale AI connectivity** (press release). The demonstration combines Lumentum's 1060 nm VCSEL optical platform, Astera Labs' Scorpio Smart Fabric Switch, and Corning's multimode fiber to extend PCIe connectivity within AI racks. The proof-of-concept targets multiple 32 Gb/s optical channels and an error-free PCIe 5.0 link, addressing the reach, cable density, power, and thermal constraints associated with electrical interconnects. Notably, Lumentum is leveraging its high-volume smartphone 3D-sensing manufacturing platform, although the announcement remains a technology demonstration rather than a commercial deployment.

Meta

- **OCP announces a switch platform standardization effort initiated by Meta, Microsoft, and Alibaba** (press release). The Open Switch Platform Architecture (OSPA) specification defines a vendor-neutral reference architecture for the switch platform management layer, extending standardization beyond the data-plane software. The initiative brings together partners including Arista, Celestica, Cisco, and Nvidia to standardize functions such as hardware inventory, power and thermal management, and firmware recovery. By enabling reusable hardware designs and software stacks, the effort aims to reduce development cycles and simplify onboarding of switches from different vendors.
- **Introduces IcePack, its first liquid-cooled networking rack, and IceCube 102.4T switch blades** (press release). IceCube delivers 128×800G non-blocking switching capacity, doubling bandwidth versus Meta's 51.2T platforms. Available in Broadcom Tomahawk6C and Nvidia Spectrum-6 variants, both run Meta's open-source FBOSS operating system and support the power and cooling requirements of gigawatt-scale AI clusters.
- **Announces Flattened Planar Fabric (FPF), a multi-plane scale-out architecture that reduces the fabric from three switch tiers to two** (press release)**.** Each GPU is connected through eight independent 100G links to eight network planes, with NIC-level hardware load balancing and host-assisted traffic steering around failed planes. The architecture supports larger non-blocking GPU clusters while reducing network hops, latency, and infrastructure cost.
- **Introduces Tahan, an Ethernet scale-up switch for MTIA 300, and previewed a 72-accelerator-per-rack architecture**. (press release). Built on Broadcom's 51.2T Tomahawk 5 ASIC, four Tahan switches interconnect 16 accelerators in a non-blocking topology, reusing Meta's scale-out RoCE transport and FBOSS software stack. The next-generation design uses eight Tomahawk6C-based switches to connect 72 accelerators per rack, with a future generation planned to adopt OCP ESUN as its scale-up data plane.

Microsoft

- **Announces Maia 300 production deployments beginning in 2027, with Maia 200 already serving production workloads** (press release). Microsoft plans to scale the Maia family to multi-gigawatt deployments starting in 2027, making it the primary inference system for its products and services. Maia 300 targets long-context reasoning and multi-step agentic workloads, with Microsoft expecting 40% more tokens per megawatt than leading GPU-based systems and reporting up to 7x Maia 200's rack-scale token throughput in a 96-accelerator configuration. Notably, Maia 200 already supports Microsoft Copilot, with GitHub Copilot and Microsoft Foundry expected to follow by year-end.

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